measured velocity depends not only on the companion's mass, but also on the mass of the star itself, and on the inclination of the binary's orbital plane to our line of sight. While the star's mass and the inclination may be inferred from careful observations, one cannot do so with ease or with good accuracy. As a result, one can readily make large errors (say, a factor of 2 or 3) in one's estimate of the mass of the dark companion. Second, black holes are not the only kind of dark companions that a star might have. For example, a neutron-star companion would also be dark. To be certain the companion is not a neutron star, one needs to be very confident that it is much heavier than the maximum allowed for a neutron star, about 2 solar masses. Two neutron stars in a tight orbit around each other could also be dark and could weigh as much as 4 Suns. The dark companion might be such a system; or it might be two cold white dwarfs in a tight orbit with total mass as much as 3 Suns. And there are other kinds of stars that, while not completely dark, can be rather massive and abnormally dim. One must look very carefully at the measured spectra to be certain there is no sign of tiny amounts of light from such stars.

Astronomers had worked hard over the preceding decades to observe and catalog binary star systems, so it was not necessary for Zel'dovich to conduct his search directly in the sky; he could search the astronomers' catalogs instead. However, he had neither the time nor the patience to comb through the catalogs himself, nor did he have the expertise to avoid all the pitfalls. Therefore, as was his custom in such a situation, he commandeered the time and the talents of someone else—in this case, Oktay Guseinov, an astronomy graduate student who already knew much about binary stars. Together, Guseinov and Zel'dovich found five promising black-hole candidates among the many hundreds of well-documented binary systems in the catalogs.

Over the next few years, astronomers paid little attention to these five black-hole candidates. I was rather annoyed at the astronomers' lack of interest, so in 1968 I enlisted Virginia Trimble, a Caltech astronomer, to help me revise and extend the Zel'dovich-Guseinov list. Trimble, though only months past her Ph.D., had already acquired a formidable knowledge of the lore of astronomy. She knew all the pitfalls we might encounter—those described above and many more—and she could gauge them accurately. By searching through the catalogs ourselves, and by collating all the published data we could find on the most promising binaries, we came